

Joining in Snowflake

INTRODUCTION TO SNOWFLAKE SQL



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JOINS

- INNER JOIN
- OUTER JOINS
 - LEFT OUTER JOIN or LEFT JOIN
 - RIGHT OUTER JOIN or RIGHT JOIN
 - FULL OUTER JOIN or FULL JOIN
- CROSS JOINS
- SELF JOINS
- NATURAL JOIN
- LATERAL JOIN

Pizza dataset



NATURAL JOIN

- **NATURAL JOIN** automatically match columns and eliminate duplicated ones

Syntax:

```
SELECT ...  
FROM <table_one> [  
    {  
        | NATURAL [ { LEFT | RIGHT | FULL } [ OUTER ] ]  
    }  
]  
JOIN <table_two>  
  
[ ... ]
```

NATURAL JOIN

Without NATURAL JOIN

```
SELECT *
FROM pizzas AS p
JOIN pizza_type AS t
ON t.pizza_type_id = p.pizza_type_id
```

PIZZA_ID	PIZZA_TYPE_ID	PIZZA_SIZE	PRICE	PIZZA_TYPE_ID_2
bbq_ckn_s	bbq_ckn	S	12.75	bbq_ckn
bbq_ckn_m	bbq_ckn	M	16.75	bbq_ckn
bbq_ckn_l	bbq_ckn	L	20.75	bbq_ckn
cali_ckn_s	cali_ckn	S	12.75	cali_ckn
cali_ckn_m	cali_ckn	M	16.75	cali_ckn

With NATURAL JOIN

```
SELECT *
FROM pizzas AS p
NATURAL JOIN pizza_type AS t
```

PIZZA_TYPE_ID	PIZZA_ID	PIZZA_SIZE	PRICE	NAME
bbq_ckn	bbq_ckn_s	S	12.75	The Barbecue Chicken Pizza
bbq_ckn	bbq_ckn_m	M	16.75	The Barbecue Chicken Pizza
bbq_ckn	bbq_ckn_l	L	20.75	The Barbecue Chicken Pizza
cali_ckn	cali_ckn_s	S	12.75	The California Chicken Pizza
cali_ckn	cali_ckn_m	M	16.75	The California Chicken Pizza

NATURAL JOIN

NOT ALLOWED

```
select *  
FROM pizzas AS p  
NATURAL JOIN pizza_type AS t  
    ON t.pizza_type_id = p.pizza_type_id
```



Syntax error: unexpected 'ON'.

NATURAL JOIN

ALLOWED

- `WHERE` clause

```
SELECT *  
FROM pizzas AS p  
NATURAL JOIN pizza_type AS t  
WHERE pizza_type_id = 'bbq_ckn'
```


LATERAL JOIN

- `LATERAL JOIN` : lets a subquery in `FROM` reference columns from preceding tables or views.

Syntax:

```
SELECT ...  
FROM <left_hand_expression> , --  
LATERAL  
(<right_hand_expression>)
```

- `left_hand_expression` - Table, view, or subquery
- `right_hand_expression` - Inline view or subquery

LATERAL JOIN with a subquery

```
SELECT
    p.pizza_id,
    lat.name,
    lat.category
FROM pizzas AS p,
LATERAL -- Keyword LATERAL
    ( SELECT *
      FROM pizza_type AS t
      -- Referencing outer query column: p.pizza_type_id
      WHERE p.pizza_type_id = t.pizza_type_id
    ) AS lat
```

Why LATERAL JOIN?

```
SELECT
    *
FROM orders AS o,
LATERAL (
    -- Subquery calculating total_spent
    SELECT
        SUM(p.price * od.quantity) AS total_spent
    FROM order_details AS od
    JOIN pizzas AS p
        ON od.pizza_id = p.pizza_id
    WHERE o.order_id = od.order_id
) AS t
ORDER BY o.order_id
```

Let's practice!

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Subquerying and Common Table Expressions

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Subquerying

- Nested queries
- Used in `FROM` , `WHERE` , `HAVING` or `SELECT` clauses
- Example:

```
SELECT column1  
FROM table1  
WHERE column1 = (SELECT column2 FROM table2 WHERE condition)
```

- Types: Correlated and uncorrelated subqueries

Uncorrelated subquery

```
-- Main query returns pizzas priced at the maximum value found in the subquery
SELECT pizza_id
FROM pizzas
-- Uncorrelated subquery that identifies the highest pizza price
WHERE price = (
    SELECT MAX(price)
    FROM pizzas
)
```

- Subquery doesn't interact with the main query

Correlated subquery

- Subquery references columns from the main query

```
SELECT pt.name,  
       pz.price,  
       pt.category  
FROM pizzas AS pz  
JOIN pizza_type AS pt  
  ON pz.pizza_type_id = pt.pizza_type_id  
WHERE pz.price = (  
  -- Identifies max price for each pizza category  
  SELECT MAX(p2.price) -- Max price  
  FROM pizzas AS p2  
  WHERE -- Correlated: uses outer query column  
        p2.pizza_type_id = pz.pizza_type_id  
)
```


Common Table Expressions

General Syntax:

```
-- WITH keyword
WITH cte1 AS ( -- CTE name
    SELECT col_1, col_2
    FROM table1
)
...
SELECT ...
FROM cte1 -- Query CTE
;
```


Common Table Expressions

```
WITH max_price AS ( -- CTE called max_price
    SELECT pizza_type_id,
           MAX(price) AS max_price
    FROM pizzas
    GROUP BY pizza_type_id
)
-- Main query
SELECT pt.name,
       pz.price,
       pt.category
FROM pizzas AS pz
JOIN pizza_type AS pt ON pz.pizza_type_id = pt.pizza_type_id
JOIN max_price AS mp -- Joining with CTE max_price
    ON pt.pizza_type_id = mp.pizza_type_id
WHERE pz.price < mp.max_price -- Compare the price with max_price CTE column
```

Multiple CTEs

```
-- Define multiple CTEs separated by commas
WITH cte1 AS (
    SELECT ...
    FROM ...
),
cte2 AS (
    SELECT ...
    FROM ...
)
-- Main query combining both CTEs
SELECT ...
FROM cte1
JOIN cte2 ON ...
WHERE ...
```

Why Use CTEs?

- Managing complex operations
- Modular
- Readable
- Reusable

Let's practice!

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Snowflake Query Optimization

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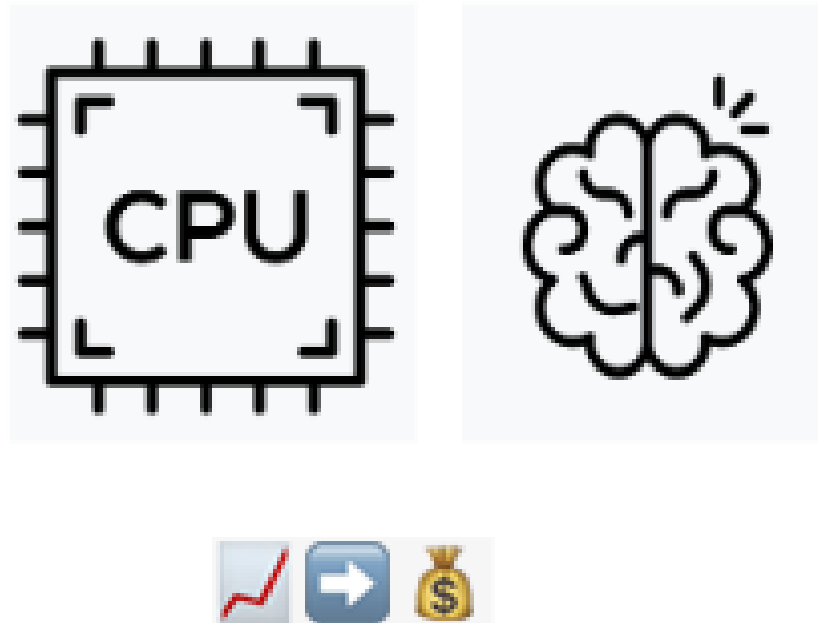


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Why Optimize Queries in Snowflake?

- Achieve faster results
- Cost efficiency
 - Shorter query times consumes fewer resources like CPU and memory.



Common query problems

- Exploding Joins: Be cautious!

Incorrect

```
SELECT *  
FROM order_details AS od  
JOIN pizzas AS p -- Missing ON condition leading to exploding joins
```

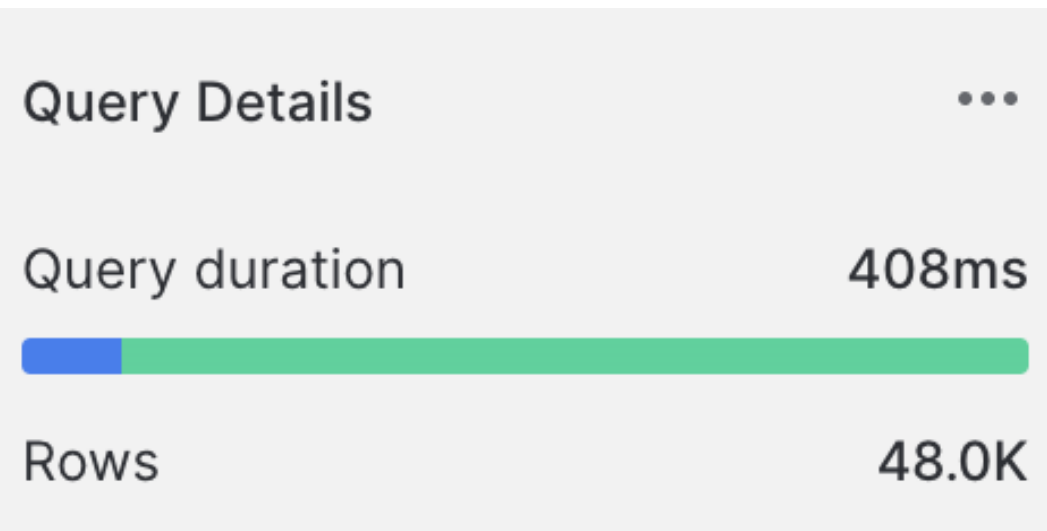
Query Details	...
Query duration	7.1s
	
Rows	4.6M

Common query problems

- Exploding Joins: Be cautious!

Correct

```
SELECT *  
FROM order_details AS od  
JOIN pizzas AS p  
ON od.pizza_id = p.pizza_id
```



Common query problems

- `UNION` or `UNION ALL` : Know the difference
 - `UNION` removes duplicates, slows down the query
 - `UNION ALL` is faster if no duplicates
- Handling big data
 - Use filters to narrow down data
 - Apply limits for quicker results

How to optimize queries?

SELECT *

```
SELECT
  *
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF100.ORDERS
```

Query Details ...

Query duration 1m 51s



Rows 150M

SELECT LIMIT 10* ⚡

```
SELECT *
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF100.ORDERS
LIMIT 10
```

Query Details ...

Query duration 1.0s



Rows 10

How to optimize queries?

Using `SELECT *`

```
SELECT
  *
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF100.ORDERS
```

Query Details ...

Query duration 1m 51s



Rows 150M

Avoid `SELECT *` ⚡

```
SELECT o_orderdate,
       o_orderstatus
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF100.ORDERS
```

Query Details ...

Query duration 16s



Rows 150M

How to optimize queries?

Filter Early

- Use `WHERE` Clause Early On
- Apply filters before `JOIN` s
 - `JOIN` will process fewer rows

Without early filtering

```
SELECT orders.order_id,  
       orders.order_date,  
       pizza_type.name,  
       pizzas.pizza_size  
FROM orders  
JOIN order_details  
ON orders.order_id = order_details.order_id  
JOIN pizzas  
ON order_details.pizza_id = pizzas.pizza_id  
JOIN pizza_type  
ON pizzas.pizza_type_id = pizza_type.pizza_type_id  
WHERE orders.order_date = '2015-01-01';  -- Filtering after JOIN
```

With early filtering

```
WITH filtered_orders AS (  
  SELECT *  
  FROM orders  
  WHERE order_date = '2015-01-01' -- Filtering in CTE before JOIN  
)  
SELECT filtered_orders.order_id,  
       filtered_orders.order_date,  
       pizza_type.name,  
       pizzas.pizza_size  
FROM filtered_orders -- Joining with CTE  
JOIN order_details  
ON filtered_orders.order_id = order_details.order_id  
JOIN pizzas  
ON order_details.pizza_id = pizzas.pizza_id
```

Query history

- Query History
 - `snowflake.account_usage.query_history`
 - Query History provides different metrics such as execution time

```
SELECT query_text, start_time, end_time, execution_time
FROM
    snowflake.account_usage.query_history
WHERE query_text ILIKE '%order_details%'
```

QUERY_TEXT	...	START_TIME	END_TIME	EXECUTION_TIME
SELECT * FROM order_details AS od JOIN pizzas AS p ON od.pizza_id = p.pizza_id		2023-09-01 03:44:37.233 -0700	2023-09-01 03:44:38.309 -0700	529
SELECT * FROM order_details AS od JOIN pizzas AS p;		2023-09-01 03:43:37.899 -0700	2023-09-01 03:43:47.369 -0700	8,747

- `ILIKE` : Case-insensitive string-matching

Query history

- Spot slow or frequently running queries

```
SELECT query_text,  
       start_time,  
       end_time,  
       execution_time  
FROM  
  snowflake.account_usage.query_history  
WHERE  
  execution_time > 1000
```

QUERY_TEXT ...	START_TIME	END_TIME	EXECUTION_TIME
select * from customer	2023-08-08 10:21:24.128 -0700	2023-08-08 10:21:53.826 -0700	29,176

Let's practice!

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Handling semi-structured data

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Structured versus semi-structured

Example of structured data

cust_id	cust_name	cust_age	cust_email
1	cust1	40	cust1***@gmail.com
2	cust2	35	cust2***@gmail.com
3	cust3	42	cust3***@gmail.com

Example of semi-structured data

```
{
  "cust_id": 1,
  "cust_name": "cust1",
  "cust_age": 40,
  "cust_email": "cust1***@gmail.com"
},
{
  "cust_id": 2,
  "cust_name": "cust2",
  "cust_age": 35,
  "cust_email": [
    "cust2***@gmail.com",
    "cust2_alternate***@gmail.com"
  ]
}
```

Introducing JSON

- JavaScript Object Notation
- Common use cases: Web APIs and Config files
- JSON data structure:
 - Key-Value Pairs, e.g., `cust_id: 1`

```
{  
  "cust_id": 1,  
  "cust_name": "cust1",  
  "cust_age": 40,  
  "cust_email": "cust1***@gmail.com"  
}
```

JSON in Snowflake

- Native JSON support
- Flexible for evolving schemas

Comparisons:

- Postgres: Uses `JSONB`
- Snowflake: Uses `VARIANT`

How Snowflake stores JSON data

- `VARIANT` supports OBJECT and ARRAY data types
 - OBJECT: `{ "key": "value" }`
 - ARRAY: `[ "list", "of", "values" ]`
- Creating a Snowflake Table to handle JSON data

```
CREATE TABLE cust_info_json_data (  
  customer_id INT,  
  customer_info VARIANT -- VARIANT data type  
);
```

Semi-structured data functions

- `PARSE_JSON`
 - `expr` : JSON data in string format
 - Returns: `VARIANT` type, valid JSON object

PARSE_JSON

Example:

```
SELECT PARSE_JSON(  
  -- Enclosed in strings  
  '{  
    "cust_id": 1,  
    "cust_name": "cust1",  
    "cust_age": 40,  
    "cust_email": "cust1***@gmail.com"  
  }'  
  -- Enclosed in strings  
) AS customer_info_json
```

CUSTOMER_INFO_JSON
{ "cust_age": 40, "cust_email": "cust1***@gmail.com", "cust_id": 1, "cust_name": "cust1" }

OBJECT_CONSTRUCT

- OBJECT_CONSTRUCT
 - Syntax: OBJECT_CONSTRUCT([<key1>, <value1> [, <keyN>, <valueN> ...]])
 - Returns: JSON object

```
SELECT OBJECT_CONSTRUCT(  
  -- Comma separated values rather than : notation  
  'cust_id', 1,  
  'cust_name', 'cust1',  
  'cust_age', 40,  
  'cust_email', 'cust1***@gmail.com'  
)
```

ID	CUSTOMER_INFO
2	{ "cust_age": 40, "cust_email": "cust1***@gmail.com", "cust_id": 1, "cust_name": "cust1" }

Querying JSON data in Snowflake

Simple JSON

- :

SELECT

```
customer_info:cust_age, -- Use colon to access cust_age from column  
customer_info:cust_name,  
customer_info:cust_email,
```

FROM

```
cust_info_json_data;
```

CUSTOMER_INFO:CUST_AGE	CUSTOMER_INFO:CUST_NAME	CUSTOMER_INFO:CUST_EMAIL
40	"cust1"	"cust1***@gmail.com"

Querying nested JSON Data in Snowflake

Example of nested JSON

```
{
  "cust_id": 1,
  "cust_name": "cust1",
  "cust_age": 40,
  "cust_email": "cust1***@gmail.com",
  "address": {
    "street": "St1",
    "city": "Cityv",
    "state": "Statev"
  }
}
```

- Colon: `:`
- Dot: `.`

Querying nested JSON using colon/dot notations

Accessing values using colon notation

```
<column>:<level1_element>:  
<level2_element>:<level3_element>
```

SELECT

```
customer_info:address:street AS street_name
```

FROM

```
cust_info_json_data
```

STREET_NAME
"St1"

Accessing values using dot notation

```
<column>:<level1_element>.  
<level2_element>.<level3_element>
```

SELECT

```
customer_info:address.street AS street_name
```

FROM

```
cust_info_json_data
```

STREET_NAME
"St1"

Let's practice!

INTRODUCTION TO SNOWFLAKE SQL

Wrap-up

INTRODUCTION TO SNOWFLAKE SQL



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Chapter 1: Snowflake SQL and key concepts

Connecting to Snowflake

- WEB UI
- Drivers & Connectors
- Snowflake CLI

Chapter 1: Snowflake SQL and key concepts

- VARCHAR
- NUMBER
- TIMESTAMP_LTZ

Data Type conversion - What, Why, How?

Conversion Functions: TO_VARCHAR , TO_DATE

STRING functions: CONCAT , INITCAP

DATE & TIME functions: CURRENT_DATE ,
CURRENT_TIME

EXTRACT functions: GROUP BY ALL

Chapter 2: Advance Snowflake SQL Concepts

JOINS

- NATURAL JOIN
- LATERAL JOIN

Subquerying

CTEs

Chapter 2: Advance Snowflake SQL Concepts

Snowflake Query Optimization

- Common query problems: Exploding Joins, `UNION` vs `UNION ALL`
- Rewriting queries: `TOP` , `LIMIT` , Early filtering, Avoid `Select *`

Semi structured data

- `PARSE_JSON` , `OBJECT_CONSTRUCT`
- Querying JSON data in Snowflake

Is this all?

- Much more to unfold

Not addressed

- Setting context
- Roles, Users
- Setting up Virtual Warehouses
- Window functions
- Query profiling
- Materialized Views
- Clustering
- ...

Useful resources

- Snowflake documentation: <https://docs.snowflake.com/>
- Snowflake forums: <https://community.snowflake.com/s/forum>
- [Introduction to Data Modeling in Snowflake](#)
- [Snowflake Tutorial](#)

This is just the beginning!

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